

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

CE-5005(4)-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Advanced Fluid Mechanics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume missing data suitably.

1. a) How do you estimate hydropower potential? 7
b) Discuss various classifications of different types of hydropower plants. 7
2. a) Distinguish the differences between the storage and Pondage? 7
b) Define and describe load factor and utilization factor. 7
3. a) Differentiate between: 7
i) Impulse and Reaction Turbine
ii) Radial and Axial flow Turbines
iii) Inward and Outward Radial flow turbines
b) A centrifugal pump is to discharge $0.118\text{m}^3/\text{sec}$ at a speed of 1450 rpm against a head of 25m. The impeller diameter at outlet is 250mm and its width at outlet is 50mm and manometric efficiency is 75%. Determine vane angle at outer periphery of the impeller. 7
4. a) Explain advantages and disadvantages of pumped storage power plants. 7
b) Explain with neat sketches types of pumped storage plants. 7

CE-5005(4)-CBGS

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[2]

5. a) Draw and discuss characteristic curves of a pump. 7
b) A double acting reciprocating pump having piston area 0.1m has a stroke of 0.30m long. The pump is discharging 2.4 m^3 of water per minute at 45 rpm through a height of 10m. Find the slip of the pump and power required to drive the pump. 7
6. a) A centrifugal pump discharge 560 liters of water per second has to develop a head of 10 meters, the speed of rotation of the impeller being 700rpm. The manometric efficiency is 85% and the loss of head in the pump due to friction is $0.025 V^2$ 12 meters of water, where V is the velocity with which the water leaves the impeller. Assume that the velocity of flow through the impeller is constant at 2.50 meters per second and that there is no velocity of whirl at inlet.
Determine: 7
i) The diameter of the impeller
ii) The outlet area
iii) The vane angle at the outlet edge of the impeller
b) A single acting reciprocating pump has a plunger of diameter 0.3m and stroke of length 0.4m. If the speed of the pump is 60 rpm and coefficient of discharge is 0.97, determine the percentage slip and actual discharge of the pump. 7
7. a) What is priming of centrifugal pump? Why it is necessary? 7
b) Obtain the expression for minimum starting speed of centrifugal pump. 7
8. a) Explain Manometric efficiency, mechanical efficiency and overall efficiency of a centrifugal pump. 7
b) Write briefly the differences between single acting reciprocating pump and double acting multistage pump. 7

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